

EROLLA RISHVIN REDDY

Applied AI Engineer • AI Automation • Intelligent Systems

Python • Machine Learning • LLM APIs • n8n • Computer Vision • FastAPI • PostgreSQL • AI Agents

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🐙 GitHub

🌐 Portfolio

🔗 LinkedIn

Professional Summary

Computer Science Engineering student specializing in **Blockchain, IoT & Cybersecurity (BIC)** with a focused interest in **Applied Artificial Intelligence, AI Automation, Machine Learning, Generative AI, Computer Vision, and Intelligent Systems**. Experienced in engineering **AI-powered applications, LLM-integrated workflows, predictive analytics systems, computer vision pipelines, AI agents, and event-driven automation** using **Python, Google Gemini, OpenRouter, n8n, FastAPI, XGBoost, OpenCV, MediaPipe, PostgreSQL, REST APIs, and Docker**. Founder of **Rishvin Labs**, developing practical AI solutions across **cybersecurity, business process automation, IoT, intelligent retail, developer productivity, and real-time decision systems**. Hands-on experience designing end-to-end AI workflows spanning **data ingestion, API orchestration, AI inference, structured outputs, decision logic, database persistence, automation, and monitoring**. Strong foundation in **machine learning, data structures, algorithms, statistics, software engineering, system design, and secure application development**.

Education

Woxsen University

B.Tech in Computer Science and Engineering

Specialization: Blockchain, IoT & Cybersecurity (BIC)

Hyderabad, India

2024 – 2028

CGPA: **9.01/10**

AI / ML & Automation Skills

Artificial Intelligence & Machine Learning: Applied AI, Machine Learning, Predictive Analytics, AI-Assisted Decision Systems, Classification, Risk Scoring, Feature Engineering, Model Integration

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, Google Gemini API, OpenRouter, LLM API Integration, Prompt Engineering, Structured Outputs, Conversational AI, AI Agents

AI Automation & Agents: n8n, Workflow Automation, AI Agent Workflows, Event-Driven Automation, API Orchestration, Webhooks, Lead Qualification, CRM Automation, Content Automation, Human-in-the-Loop Workflows

Computer Vision: OpenCV, MediaPipe, Face Mesh, Facial Landmark Detection, Liveness Detection, Gesture Recognition, Computer Vision Pipelines, Image Processing, Product Recognition

AI Engineering & Backend: Python, FastAPI, Node.js, TypeScript, REST APIs, PostgreSQL, MongoDB, Docker, Linux, JSON, API Integration, Backend Services

Data & Analytics: SQL, Pandas, Data Processing, Data Analysis, Feature Engineering, Predictive Analytics, Real-Time Analytics, KPI Monitoring, Data Visualization

Intelligent Systems: Recommendation Systems, Intelligent Automation, Decision Support Systems, Route Optimization, AI-Powered Security Analysis, AI + IoT Systems, Real-Time Intelligent Applications

Developer Tools & Platforms: Git, GitHub, GitHub Actions, Postman, Docker, Vercel, Arduino IDE, Antigravity IDE, Overleaf

Software Engineering Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, System Design, Software Engineering, Secure Application Development

Technical Experience

Founder & Applied AI / Automation Engineer – Rishvin Labs

2025 – Present

- Founded and lead **Rishvin Labs**, developing applied AI systems and automation solutions across **Generative AI, machine learning, intelligent workflows, cybersecurity, computer vision, IoT, and full-stack engineering**.
- Architected and developed AI-powered applications integrating **LLM APIs, Google Gemini, OpenRouter, n8n, machine learning models, computer vision pipelines, REST APIs, webhooks, databases, and event-driven workflows**.
- Built automated systems for **lead qualification, conversational AI, security analysis, content generation, developer productivity, intelligent decision support, and real-time operational workflows**.
- Engineered production-oriented AI workflows with **API orchestration, structured AI outputs, idempotent processing, database persistence, error handling, automation triggers, notifications, and external service integrations**.

- Applied AI engineering across **cybersecurity, IoT, business automation, computer vision, and developer tooling**, focusing on reliable, scalable, and practical intelligent systems rather than standalone model experimentation.

Pega Platform Intern – Pegasystems

Aug 2026 – Present

- Selected for the **National Internship Program 2026 with SmartBridge**, gaining hands-on experience with enterprise application development and workflow automation using the **Pega Platform**.
- Developing practical knowledge of **Business Process Management (BPM), Case Management, Digital Process Automation (DPA), workflow orchestration, and low-code application development**.
- Applying structured enterprise development practices involving **business process modeling, case lifecycle management, application configuration, data integration, and automated business workflows**.

Applied AI & Intelligent Systems Engineering

Academic & Independent Engineering

- Designed and implemented AI-powered systems combining **machine learning, LLMs, computer vision, workflow automation, IoT data, and backend services** to solve practical engineering and business problems.
- Developed intelligent pipelines following the **event → data processing → AI inference → decision → automation** architecture across multiple independent projects.
- Integrated **predictive models, AI APIs, computer vision algorithms, webhooks, databases, and third-party APIs** into end-to-end applications and automated workflows.
- Applied software engineering principles including **system architecture, API design, database engineering, testing, debugging, observability, security, and deployment** to AI-enabled applications.

Featured AI Projects

AI Security Guardian – AI-Powered Security Operations Platform Python, n8n, PostgreSQL, Docker, LLM APIs, OWASP

- Architected an **AI-powered Security Operations Center (SOC)** automating attack surface management, vulnerability assessment, threat intelligence, security analysis, and incident workflows.
- Integrated **LLM-powered security analysis, deterministic risk scoring, automated vulnerability workflows, threat intelligence enrichment, and AI-assisted reporting** through n8n orchestration.
- Designed a containerized architecture using **Python, PostgreSQL, Docker, and event-driven automation** for repeatable security analysis and scalable AI-assisted operations.

AI Speed-to-Lead – Intelligent Lead Qualification & Voice Automation n8n, LLM APIs, Voice AI, Webhooks, PostgreSQL, CRM Automation

- Engineered an **AI-powered lead qualification and voice automation system** for real-time lead capture, identity resolution, automated outreach, and CRM lifecycle management.
- Implemented **AI voice interactions, secure webhooks, transcript intelligence, lead scoring, qualification logic, opt-out enforcement, and high-intent notifications**.
- Designed an event-driven architecture with **idempotent processing, API orchestration, database persistence, automated decision workflows, and real-time business notifications**.

EcoBin – AI-Powered Smart Waste Management System Python, FastAPI, React, TypeScript, XGBoost, Google OR-Tools

- Developed an **AI-powered intelligent waste management platform** combining IoT telemetry, predictive analytics, machine learning, and automated collection planning.
- Built predictive services using **XGBoost and FastAPI** for container fill-level prediction and overflow-risk analysis, with a React-based operational dashboard.
- Applied **Google OR-Tools** for route optimization, connecting machine-learning predictions with automated operational decision-making.

Smart Cart OS – AI-Powered Intelligent Shopping System Python, TypeScript, React, RFID, Computer Vision, Gemini AI

- Designed an **AI-enabled smart shopping system** combining RFID-based product identification, real-time analytics, computer vision, and conversational AI capabilities.
- Integrated **Google Gemini AI and computer vision** for automated product recognition and intelligent shopping workflows.
- Developed a connected application architecture supporting **real-time data processing, AI inference, product intelligence, and interactive user experiences**.

Face-Mesh Verification – Real-Time Computer Vision System Python, OpenCV, MediaPipe, Flask, Computer Vision

- Built a **real-time computer vision platform** using MediaPipe Face Mesh for facial landmark detection, liveness verification, and human-computer interaction.
- Implemented **468-point facial landmark tracking, blink-gated liveness detection, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and facial feature analysis**.

- Extended the vision pipeline with **nose-tip cursor control and real-time interaction**, demonstrating practical computer vision and AI-assisted interface engineering.

GitHub-to-LinkedIn AI Publisher – Automated Developer Content Pipeline n8n, OpenRouter, GitHub API, LinkedIn API, Python

- Built an **AI-powered content automation pipeline** that transforms GitHub activity into structured, professional LinkedIn content with minimal manual intervention.
- Integrated **GitHub events, OpenRouter LLM APIs, content generation, workflow orchestration, and LinkedIn publishing** into an end-to-end automated pipeline.
- Implemented event-driven processing to **detect development activity, generate context-aware content, validate outputs, and automate social publishing workflows**.

Certifications

Programming for Everybody (Getting Started with Python) – University of Michigan (Coursera)

Data Structures Using Python: An Introduction – Coursera

Career Essentials in GitHub Professional Certificate – GitHub & LinkedIn Learning

Practical GitHub Actions – LinkedIn Learning

Practical GitHub Code Search – LinkedIn Learning

Practical GitHub Project Management and Collaboration – LinkedIn Learning

Introduction to Cyber Attacks – New York University (Coursera)

Object-Oriented Design – University of Alberta (Coursera)

Java Programming – Coursera

Relevant Coursework

AI & Data Foundations: Probability & Statistics, Data Structures & Algorithms, Database Management Systems, Data Processing, Problem Solving

Software Engineering: Object-Oriented Programming, Software Engineering, System Design, Application Development, Database Design

Computing Foundations: Computer Networks, Operating Systems, Internet Technologies, Computer Architecture

Supporting Domains: Cybersecurity, Internet of Things, Blockchain, Technical Writing

Patent & Innovation

Government of India Registered Design Patent – IoT Connectivity Device *Co-Inventor — Design No: 470097-001*

- Co-inventor of a **Government of India registered design patent** for an IoT connectivity device, demonstrating applied innovation across **connected systems, intelligent infrastructure, and device communication**.
- Contributed to an engineering solution focused on **IoT connectivity and device integration**, demonstrating the ability to translate an engineering concept into a formally registered intellectual property asset.